

Activity 24 Instruction Manual

Activity 24 | Topic 03 | C013 | Version v1.0

Purpose

Quality control is about detecting problems from data rather than opinion. You will collect data on a check sheet and plot a control chart, applying the rule of seven.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A check sheet of collected data and a control chart with limits and a trend assessment.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. In your group, open 'Check Sheets.xlsx'. Prompt GenAI: 'Design a check sheet to collect defect data for [your project's deliverable]. Suggest the defect categories and the collection frequency.'
2. Confirm the categories are mutually exclusive - a defect must fall into exactly one. Fix any overlap the AI created.
3. Populate the check sheet with 20 data points, either real data or a plausible set your group agrees on.
4. Open 'Control Chart.xlsx'. Enter your data and compute the mean, the upper control limit and the lower control limit.
5. Understand the distinction: control limits come from the process itself (what it actually does); specification limits come from the customer (what they require). A process can be in control and still fail the specification.
6. Apply the rule of seven: seven consecutive points on the same side of the mean, or seven consecutively increasing or decreasing, indicates a trend requiring investigation - even if no point breaches a control limit.
7. Prompt GenAI: 'Analyse this control chart data. Is the process in control? Are there any runs of seven? What would you investigate?' Verify its answer against your own reading of the chart.
8. Follow the full control quality sequence: check sheet → histogram → Pareto → cause-and-effect → scatter analysis. State which step you would do next and why.
9. Share whether your process is in control and what you would investigate.

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- Your control chart has a mean and both control limits, you have checked for runs of seven, and you can distinguish control limits from specification limits.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.